$$A e^{-at} \underbrace{\left(e^{\Delta t} - 1\right)}_{g(t)} \; ; \; \Delta > 0$$

$$\frac{g'}{g} = a$$

optimum at :

$$= e^{-at} g(t)$$

opt time at

$$-a g(t) + g'(t) = 0$$

$$R(t) = e^{-a(t-t_*)} \frac{g(t)}{g(t^*)}$$

$$x = a(t_* - t)$$

$$t = t^* - \frac{x}{a}$$

$$= e^{a(t^* - t)} \frac{g\left(t^* - \frac{x}{a}\right)}{g(t^*)} \approx e^{x} \left(1 - \frac{x}{a} \frac{g'}{g}\right)$$

1

nonon (not Ichar)

lity rate

$$\dot{X} = -aX \qquad X(0) = X_0$$

$$\dot{Y} = aX - bY \quad , Y(0) = 0$$

solution:

$$X(t) = X_0 e^{-at}$$

$$Y(t) = \frac{X_0}{b-a}\left(e^{-at} - e^{-bt}\right)$$

$$= \frac{X_0}{a-b}\left(e^{-bt} - e^{-at}\right)$$

a-b

$\frac{1}{a-b}$

nonon (not Chac)

lity rate

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$$= \frac{X_0}{a-b}\left(e^{-bt} - e^{-at}\right)$$

$b > a$ $\quad$ $a > b$

$\frac{a}{a-b}$ $\quad$ $\frac{b}{a-b}$